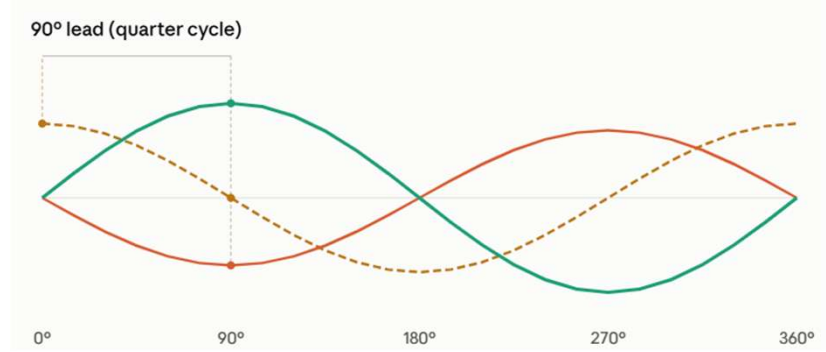
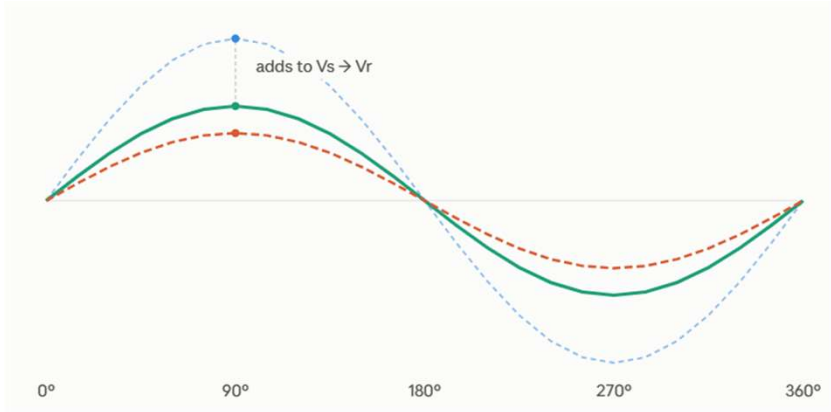


Vs — sending-end voltage (base, 0°)

Ic — charging current, leads Vs by 90°

Ic × XL — the drop, antiphase with Vs

At 90°: Vs peaks while Ic×XL is at its most negative.

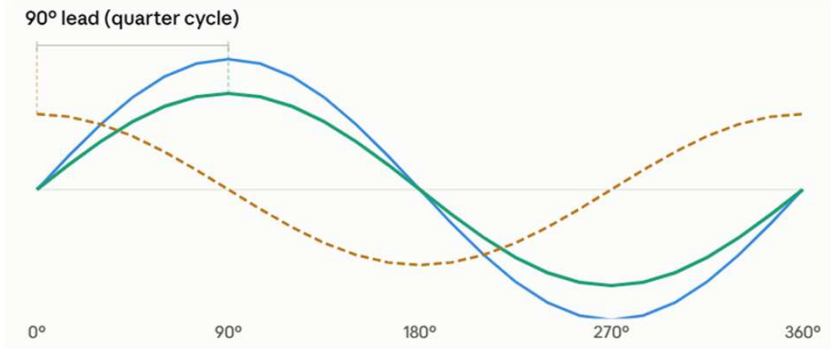


Vs — sending-end voltage (base, 0°)

−Ic×XL — now in phase with Vs

Vr — the sum, same phase, larger peak

Vs and −Ic×XL peak together — they add, not cancel.



Vs — sending-end voltage (base reference, 0°)

Vr — receiving-end voltage, same phase, larger

Ic — charging current, leads by 90°

Vr and Vs peak together — only the amplitude differs.